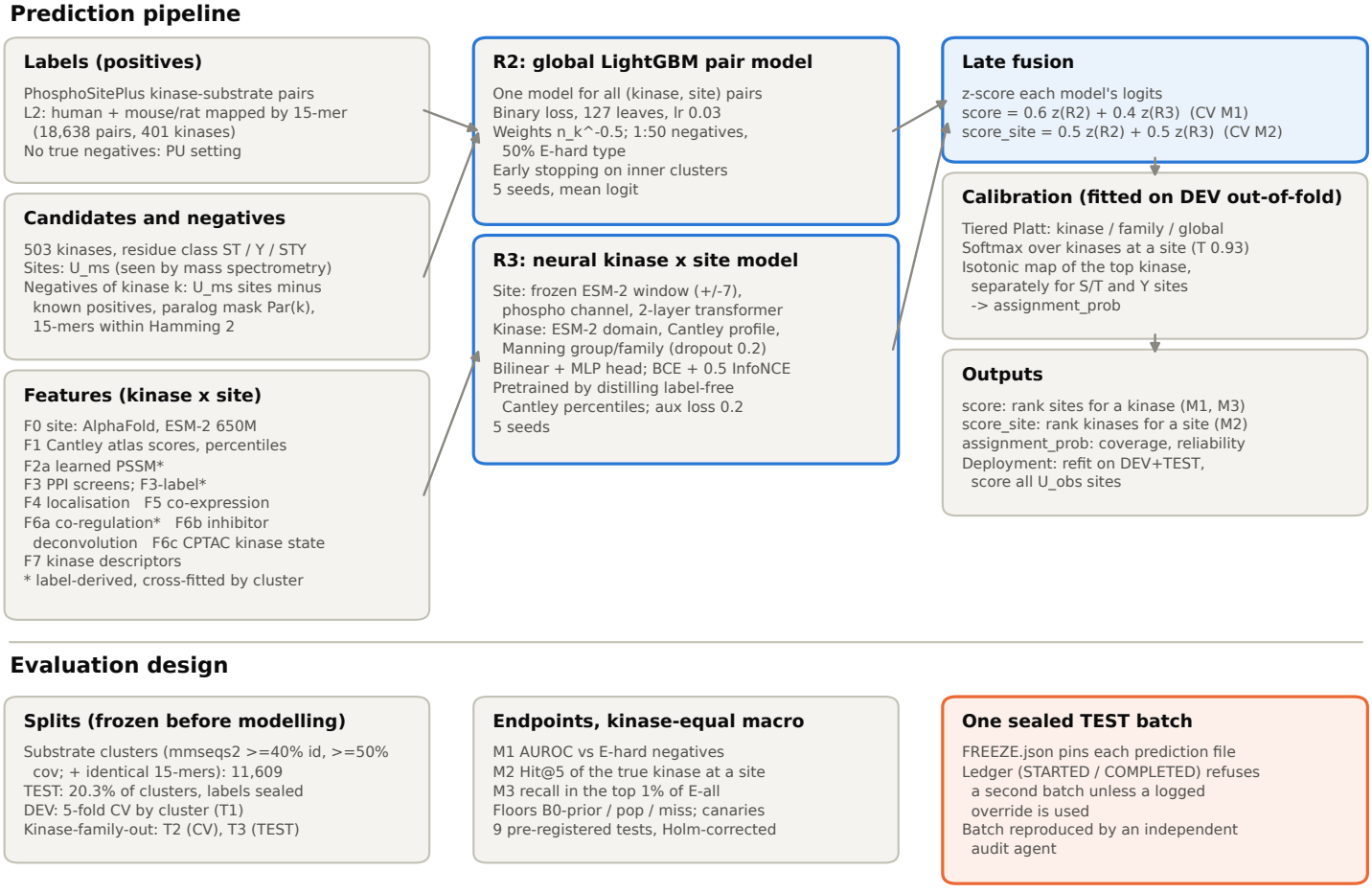
Prediction pipeline
Labels (positives)
PhosphoSitePlus kinase-substrate pairs
L2: human + mouse/rat mapped by 15-mer
(18,638 pairs, 401 kinases)
No true negatives: PU setting
Candidates and negatives
503 kinases, residue class ST / Y / STY
Sites: U_ms (seen by mass spectrometry)
Negatives of kinase k: U_ms sites minus
known positives, paralog mask Par(k),
15-mers within Hamming 2
Features (kinase x site)
F0 site: AlphaFold, ESM-2 650M
F1 Cantley atlas scores, percentiles
F2a learned PSSM*
F3 PPI screens; F3-label*
F4 localisation F5 co-expression
F6a co-regulation* F6b inhibitor
deconvolution F6c CPTAC kinase state
F7 kinase descriptors
* label-derived, cross-fitted by cluster
R2: global LightGBM pair model
One model for all (kinase, site) pairs
Binary loss, 127 leaves, lr 0.03
Weights n_k^-0.5; 1:50 negatives,
50% E-hard type
Early stopping on inner clusters
5 seeds, mean logit
R3: neural kinase x site model
Site: frozen ESM-2 window (+/-7),
phospho channel, 2-layer transformer
Kinase: ESM-2 domain, Cantley profile,
Manning group/family (dropout 0.2)
Bilinear + MLP head; BCE + 0.5 InfoNCE
Pretrained by distilling label-free
Cantley percentiles; aux loss 0.2
5 seeds
Late fusion
z-score each model's logits
score = 0.6 z(R2) + 0.4 z(R3) (CV M1)
score_site = 0.5 z(R2) + 0.5 z(R3) (CV M2)
Calibration (fitted on DEV out-of-fold)
Tiered Platt: kinase / family / global
Softmax over kinases at a site (T 0.93)
Isotonic map of the top kinase,
separately for S/T and Y sites
-> assignment_prob
Outputs
score: rank sites for a kinase (M1, M3)
score_site: rank kinases for a site (M2)
assignment_prob: coverage, reliability
Deployment: refit on DEV+TEST,
score all U_obs sites
Evaluation design
Splits (frozen before modelling)
Substrate clusters (mmseqs2 >=40% id, >=50%
cov; + identical 15-mers): 11,609
TEST: 20.3% of clusters, labels sealed
DEV: 5-fold CV by cluster (T1)
Kinase-family-out: T2 (CV), T3 (TEST)
Endpoints, kinase-equal macro
M1 AUROC vs E-hard negatives
M2 Hit@5 of the true kinase at a site
M3 recall in the top 1% of E-all
Floors B0-prior / pop / miss; canaries
9 pre-registered tests, Holm-corrected
One sealed TEST batch
FREEZE.json pins each prediction file
Ledger (STARTED / COMPLETED) refuses
a second batch unless a logged
override is used
Batch reproduced by an independent
audit agent